

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396538
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	deGrood; Michael John et al.

Hair styler including hinged arms having twisted hair contact surfaces

Abstract

A hair styler includes an upper arm, a lower arm, and a hinge. The upper arm includes an upper grip and an upper clamp head. The upper clamp head includes an upper hair contact surface. The upper hair contact surface is twisted relative to the upper grip. The lower arm includes a lower grip and a lower clamp head. The lower clamp head includes a lower hair contact surface. The lower hair contact surface is twisted relative to the lower grip. The hinge attaches the upper and lower arms together such that the upper and lower arms are openable and closeable relative to one another. The hair styler is configured to receive hair between the upper and lower hair contact surfaces when the hair styler is in a closed position.

Inventors:	deGrood; Michael John (Madison, WI), Kuzia; Jay William (Madison, WI)
Applicant:	Spectrum Brands, Inc. (Middleton, WI)
Family ID:	1000008780795
Assignee:	Spectrum Brands, Inc. (Middleton, WI)
Appl. No.:	18/670374
Filed:	May 21, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240298766 A1	Sep. 12, 2024

Foreign Application Priority Data

CN	201811611231.0	Dec. 27, 2018
----	----------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **A45D1/04** (20060101); **A45D1/12** (20060101); **A45D2/00** (20060101); A45D1/18 (20060101)

U.S. Cl.:

CPC **A45D1/04** (20130101); **A45D1/12** (20130101); **A45D2/001** (20130101); A45D1/18 (20130101); A45D2/002 (20130101)

Field of Classification Search

CPC: A45D (1/02); A45D (1/04); A45D (1/06); A45D (1/08); A45D (1/10); A45D (1/14); A45D (1/18); A45D (1/28); A45D (2/001); A45D (2/002); A45D (2/40); A45D (2200/205)

USPC: D28/35; D28/37

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
866778	12/1906	Eldridge	N/A	N/A
1397332	12/1920	Russell	N/A	N/A
1491415	12/1923	Pegelow	N/A	N/A
4464562	12/1983	Takimae	N/A	N/A
5046516	12/1990	Barradas	N/A	N/A
5120933	12/1991	Altamore et al.	N/A	N/A
5711323	12/1997	Denebeim	N/A	N/A
6070596	12/1999	Altamore	132/229	A45D 1/06
6920886	12/2004	McCambridge et al.	N/A	N/A
7044139	12/2005	Tong	N/A	N/A
8082929	12/2010	Kim	N/A	N/A
8237089	12/2011	Legrain et al.	N/A	N/A
8763283	12/2013	Bearley	38/77.83	D06F 71/34
8955526	12/2014	Smith	N/A	N/A
9173465	12/2014	Morgandi	N/A	A45D 1/04
D747038	12/2015	Speh	N/A	N/A
D776871	12/2016	Smith	N/A	N/A
D791408	12/2016	Brady	D28/35	N/A
10092076	12/2017	Mandica et al.	N/A	N/A
D875312	12/2019	Hu	D28/35	N/A
D901770	12/2019	Wang	D28/35	N/A
10912363	12/2020	Kanter	N/A	A45D 19/012
2002/0066022	12/2001	Calder et al.	N/A	N/A

2003/0000542	12/2002	Huntley	N/A	N/A
2004/0124188	12/2003	Altamore	N/A	N/A
2005/0150511	12/2004	Park	N/A	N/A
2005/0220993	12/2004	Sterzel	N/A	N/A
2006/0201527	12/2005	Higgins et al.	N/A	N/A
2008/0035167	12/2007	Chan	N/A	N/A
2008/0184510	12/2007	Conceicao	N/A	N/A
2008/0303404	12/2007	Kawanami et al.	N/A	N/A
2010/0132733	12/2009	Kyu et al.	N/A	N/A
2010/0269848	12/2009	Morgandi	132/269	A45D 1/04
2011/0061672	12/2010	Schmider	132/226	A45D 1/12
2011/0061673	12/2010	Yahnker et al.	N/A	N/A
2011/0120490	12/2010	King	N/A	N/A
2011/0220141	12/2010	Chan	132/229	A45D 1/04
2011/0232673	12/2010	Crawford	N/A	N/A
2012/0227758	12/2011	Ford	132/229	A45D 1/04
2013/0146081	12/2012	Chan	N/A	N/A
2013/0247934	12/2012	Han	132/269	A45D 2/002
2013/0255715	12/2012	Lund	132/225	A45D 2/40
2013/0263881	12/2012	Story	N/A	N/A
2014/0238432	12/2013	Deng	132/229	A45D 1/04
2014/0261524	12/2013	Smith	N/A	N/A
2015/0114425	12/2014	Park	N/A	N/A
2015/0122284	12/2014	Smith	N/A	N/A
2015/0150350	12/2014	Kock	132/229	A45D 1/04
2015/0237983	12/2014	Kock	132/229	A45D 2/001
2016/0213116	12/2015	Williams	N/A	N/A
2016/0255936	12/2015	deGrood	N/A	A45D 1/04
2016/0278496	12/2015	Smith	N/A	N/A
2016/0286928	12/2015	Weatherly	N/A	A45D 2/40
2016/0360846	12/2015	Shami	N/A	N/A
2017/0231350	12/2016	Moore	132/224	A45D 2/001
2018/0177273	12/2017	Jeong	N/A	A45D 1/08
2020/0205540	12/2019	Wong	N/A	A45D 1/12
2021/0082310	12/2020	Moore	N/A	G09B 5/04
2022/0322802	12/2021	Debenedictis	N/A	A45D 1/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
206137462	12/2016	CN	N/A
207023588	12/2017	CN	N/A
0077600	12/1982	EP	N/A
2095736	12/2008	EP	N/A
2003310339	12/2002	JP	N/A
2003339422	12/2002	JP	N/A
2009136304	12/2008	JP	N/A
2011512225	12/2010	JP	N/A
2014217748	12/2013	JP	N/A
20100001162	12/2009	KR	N/A
101311808	12/2012	KR	N/A

101564408	12/2014	KR	N/A
M515311	12/2015	TW	N/A
2015071656	12/2014	WO	N/A

OTHER PUBLICATIONS

CN207023588U (Year: 2018). cited by examiner

CN206137462U (Year: 2017). cited by examiner

Extended European Search Report for EP Patent Application 19903646.8 dated Sep. 1, 2022; 9 pp. cited by applicant

International Search Report and Written Opinion for Patent Application PCT/US2019/068569 mailed Mar. 12, 2020; 13 pp. cited by applicant

Notice of Reasons for Refusal for Japanese Patent Application No. 2019-225949 dated May 31, 2021; 6 pp. cited by applicant

GHD, "You don't always have to think straight," www.ghdhairusa.com; 1 pp. cited by applicant

Primary Examiner: Steitz; Rachel R

Assistant Examiner: Asqiriba; Karim

Attorney, Agent or Firm: Armstrong Teasdale LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/419,208 filed on Jun. 28, 2021, which is a 371 national stage of PCT Application No. PCT/US2019/068569 filed on Dec. 26, 2019, which claims the benefit of Chinese Patent Application No. 201811611231.0 filed on Dec. 27, 2018, which are incorporated by reference herein in their entirety.

BACKGROUND

(1) The present disclosure relates generally to a hair styler and, more particularly, to a hair styler including hinged arms and twisted hair contact surfaces.

(2) Many conventional hair stylers have a pair of arms that are hingedly connected together to permit manual opening and closing of the arms relative to one another for straightening or curling hair between the arms. Each arm typically includes a clamp head and a hair contact surface on the clamp head. During use, hair is positioned between the hair contact surfaces and the hair is styled. The hair contact surfaces are generally flat and straight. Accordingly, a user may have difficulty in performing some styling operations with the hair stylers. In addition, the hair stylers may require the users to hold the hair stylers in awkward orientations when styling the hair and may not accommodate the users' preferred methods for gripping and manipulating the styler because of the orientations of the straight and flat hair contact surfaces relative to the handles.

(3) There is a need, therefore, for a hair styler that can be used for a larger range of styling operations and that is more intuitive for a user to style their hair with as compared to at least some conventional hair straighteners.

SUMMARY

(4) In one aspect, a hair styler generally comprises an upper arm, a lower arm, and a hinge. The upper arm includes an upper grip and an upper clamp head. The upper clamp head includes an upper hair contact surface. The upper hair contact surface is twisted relative to the upper grip. The lower arm includes a lower grip and a lower clamp head. The lower clamp head includes a lower

hair contact surface. The lower hair contact surface is twisted relative to the lower grip. The hinge attaches the upper and lower arms together such that the upper and lower arms are openable and closeable relative to one another. The hair styler is configured to receive hair between the upper and lower hair contact surfaces when the hair styler is in a closed position.

(5) In another aspect, a hair styler generally comprises a first arm including a first clamp plate and a second arm including a second clamp plate. The first and second clamp plates are twisted. The hair styler further includes a hinge attaching the first and second arms together such that the first and second arms are openable and closeable relative to one another. The hair styler is configured to receive hair between the first and second clamp plates when the hair styler is in a closed position. The hair styler also includes a heating system configured to deliver heat to the hair positioned between the first and second clamp plates.

(6) In yet another aspect, a hair styler generally comprises a first clamp head including a first hair contact surface and a second clamp head including a second hair contact surface. The first and second hair contact surfaces are twisted. The hair styler also includes a handle attached to at least one of the first clamp head and the second clamp head, and a hinge attaching the first and second clamp heads together such that the first and second clamp heads are openable and closeable relative to one another. The hair styler is configured to receive hair between the first and second hair contact surfaces when the hair styler is in a closed position.

Description

BRIEF DESCRIPTION

(1) FIG. 1 is a perspective view of one suitable embodiment of a hair styler, the styler including upper and lower hinged arms having twisted (i.e., non-planar) hair contact surfaces, the hinged arms being in an opened position.

(2) FIG. 2 is an end view of the hair styler of FIG. 1, with the hinged arms of the appliance in a closed position.

(3) FIG. 3 is an end view of the hair styler of FIG. 1, with the hinged arms in the opened position.

(4) FIG. 4 is a side view of another suitable embodiment of a hair styler in an opened position, the hair styler including hinged upper and lower arms having twisted hair contact surfaces and teeth along edges of the surfaces.

(5) FIG. 5 is a side view of the hair styler of FIG. 4 in a closed position.

(6) FIG. 6 is a perspective view of a portion of the hair styler of FIG. 4 with the upper arm removed to illustrate the twisted hair contact surface of the lower arm of the styler.

(7) FIG. 7 is a perspective view of a portion of the hair styler of FIG. 4 with the lower arm removed to illustrate the twisted hair contact surface of the upper arm of the styler.

(8) FIG. 8 is an enlarged perspective view of a portion of the hair styler as seen in FIG. 5 illustrating the teeth along the twisted hair contact surfaces of the hinged arms of the hair styler.

(9) FIG. 9 is a cross-sectional view of the hair styler taken along section line 9-9 shown in FIG. 8.

(10) FIG. 10 is a cross-sectional view of the hair styler taken along section line 10-10 shown in FIG. 8.

(11) FIG. 11 is a perspective view of yet another embodiment of a hair styler in a closed position, the hair styler including hinged upper and lower arms having twisted hair contact surfaces and upper and lower grips on opposite ends of the arms.

(12) FIG. 12 is an enlarged perspective view of a portion of the hair styler of FIG. 11.

(13) FIG. 13 is a perspective view of a portion of the hair styler of FIG. 11 with the upper arm removed to illustrate the twisted hair contact surface of the lower arm of the styler.

(14) FIG. 14 is a perspective view of a portion of the hair styler of FIG. 11 with the lower arm removed to illustrate the twisted hair contact surface of the upper arm of the styler.

(15) Corresponding reference characters indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(16) As used herein, the terms “twist” or “twisted” refer to non-planar surfaces. For example, the non-planar surfaces may be curved or twisted relative to an axis.

(17) Referring now to the drawings, and in particular to FIGS. 1-3, a hair styler of one suitable embodiment of the present disclosure is indicated generally by the reference numeral **10**. The illustrated hair styler **10** comprises a pair of arms (namely, an upper or first arm indicated at **12** and a lower or second arm indicated at **14**), a hinge **16** by which the arms **12**, **14** are connected together, and a power cord **18** (partially illustrated in FIGS. 1-3). The hinge **16** is configured to facilitate opening and closing the arms **12**, **14** relative to one another. The hair styler **10** has a longitudinal axis **60** extending through the hinge **16**. The arms **12**, **14** extend along the longitudinal axis **60** when the arms are in the closed position.

(18) For operating the hair styler **10** in the manner set forth herein, the styler **10** may be provided with a power button (or “ON/OFF” switch), a temperature increase button (or “UP” switch), a temperature decrease button (or “DOWN” switch), and a graphical user interface (e.g., an LCD display). Suitably, in other embodiments, the hair styler **10** may be configured in any manner that facilitates user operation of the hair styler **10** as set forth herein (e.g., the hair styler **10** may have any suitable number of buttons and displays arranged in any suitable manner).

(19) The upper arm **12** includes an upper grip **20** and an upper clamp head **22**. The upper grip **20** includes an outer wall **24** and an inner plate **26** (FIG. 3). The upper clamp head **22** includes a top **30**, two sides **32**, and a clamp plate **34** opposite the top. The clamp plate **34** has an upper hair contact surface **36** that is twisted or non-planar. For example, the upper hair contact surface **36** extends between the sides **32** of the upper clamp head **22** at a first twist angle **38** relative to the inner plate **26** of the upper grip **20**. The twist angle refers to the angular measurement of the curve or twist of a surface. In some suitable embodiments, the first twist angle **38** of the upper hair contact surface **36** may be in a range of about 60° to about 120°.

(20) The lower arm **14** includes a lower grip **40** and a lower clamp head **42**. The lower grip **40** includes an outer wall **44** and an inner plate **46** (FIG. 3). The lower clamp head **42** includes a bottom **48**, two sides **50**, and a clamp plate **52** opposite the bottom. The clamp plate **52** has a lower hair contact surface **54** that extends between the sides **50** of the lower clamp head **42** at a second twist angle **56** relative to the inner plate **46** of the lower grip **40**. In some suitable embodiments, the second twist angle **56** of the lower hair contact surface **54** may be in a range of about 60° to about 120°.

(21) The first twist angle **38** of the upper hair contact surface **36** and the second twist angle **56** of the lower clamp head **42** provide a twisted configuration of the hair styler **10** that enables the hair styler to be used for a range of styling operations including straightening or curling hair. Moreover, the twisted configuration of the hair styler **10** makes the hair styler more intuitive for a user to grasp and manipulate during a styling operation and provides a clear indication to the user that the hair styler **10** may be used to curl hair.

(22) The hair styler **10** is positionable between a closed position (shown in FIG. 2) and an opened position (shown in FIGS. 1 and 3) by moving the upper arm **12** and the lower arm **14** relative to each other about the hinge **16**. In the closed position, the upper arm **12** and the lower arm **14** are configured to receive and clamp hair between the upper hair contact surface **36** and the lower hair contact surface **54**. Suitably, the first twist angle **38** is equal to the second twist angle **56** and the upper hair contact surface **36** and the lower hair contact surface **54** are configured to fit together and clamp the hair therebetween when the styler **10** is in the closed position.

(23) In addition, the upper grip **20** and the lower grip **40** are sized and shaped to enable a user to grasp the hair styler **10** and position the arms **12**, **14** in the opened and closed position.

Accordingly, the upper grip **20** and the lower grip **40** at least partly form a handle for the user to

grasp and manipulate the styler **10**. Sometimes, the user may grasp the styler **10** using either the upper grip **20** or the lower grip **40**. In other embodiments, a separate handle may be connected to the arms **12, 14**.

(24) In the illustrated embodiment, the outer wall **24** of the upper grip **20**, the outer wall **44** of the lower grip **40**, the top **30** of the upper clamp head **22**, and the bottom **48** of the lower clamp head **42** are shaped such that the hair styler **10** is generally cylindrical when the styler is in the closed position. Also, the upper hair contact surface **36** and the lower hair contact surface **54** are twisted relative to the grips **20, 40**. As a result, the user is able to comfortably grasp and manipulate the hair styler **10** and use the twisted portions of the arms **12, 14** to style hair.

(25) As shown in FIGS. **1-3**, the first twist angle **38** and the second twist angle **56** vary along the longitudinal axis **60** of the hair styler **10** to provide the twisted shape of the upper hair contact surface **36** and the lower hair contact surface **54**. In other words, the upper hair contact surface **36** and the lower hair contact surface **54** are nonplanar and are curved along the longitudinal axis **60** of the styler **10** and perpendicular to the longitudinal axis. In contrast and as best seen in FIG. **1**, the inner plate **26** of the upper grip **20** and the inner plate **46** of the lower grip **20** are generally planar and extend parallel to the longitudinal axis **60** of the styler **10** when the hair styler is in the closed position. Accordingly, the styler **10** is twisted along the portion of the arms **12, 14** that contact the hair and is not twisted along the portion of the arms **12, 14** that is grasped by the user.

(26) In some suitable embodiments, the hair styler **10** is configured to deliver heat to hair positioned between the arms **12, 14** for the styling operation. For example, the upper hair contact surface **36** and/or the lower hair contact surface **54** may be heating surfaces of heat generating plates connected to electric control devices. Accordingly, the hair styler **10** is configured to deliver heat to the hair through the twisted hair contact surfaces **36, 54**.

(27) During operation, a user opens the hair styler **10** and positions hair between the arms **12, 14**. The hair extends across and contacts the hair contact surfaces **36, 54**. Heat may be delivered to the hair through at least one of the hair contact surfaces **36, 54**. The user can then move the styler **10** along the hair to perform a styling operation such as curling or straightening the hair. The twisted shape of the styler **10** allows the user to perform a range of styling operations and grasp the styler comfortably. When the styling operation for the hair has been completed, the user opens the styler **10** and removes the hair. The user may then position another section of hair between the arms **12, 14** and clamp the styler **10** onto the hair.

(28) Referring to FIGS. **4** and **5**, a hair styler of another suitable embodiment of the present disclosure is generally indicated by the reference number **100**. The illustrated hair styler **100** comprises a pair of arms (namely, an upper or first arm indicated at **102** and a lower or second arm indicated at **104**), a hinge **106** by which the arms **102, 104** are connected together, and a power cord **108** (partially illustrated in FIGS. **4** and **5**). The hinge **106** is configured to facilitate opening and closing the arms **102, 104** relative to one another. The arm **104** is provided with a user interface **110** such as a touch screen including a power button (or “ON/OFF” switch), a temperature increase button (or “UP” switch), a temperature decrease button (or “DOWN” switch), and a graphical user interface (e.g., an LCD display). In addition, the arm **102** may include a lock **112** to lock the styler **100** in the opened or closed position. Suitably, in other embodiments, the hair styler **100** may be configured in any manner that facilitates user operation of the hair styler **100** as set forth herein (e.g., the hair styler **100** may have any suitable number of buttons and displays arranged in any suitable manner).

(29) The upper arm **102** includes an upper grip **120** and an upper clamp head **122**. The upper grip **120** includes an outer wall **124** and an inner plate **126** (FIG. **7**). The upper clamp head **122** includes a top **130**, two sides **132**, and an upper clamp plate **134**. The upper clamp plate **134** has an upper hair contact surface **136** that extends between the sides **132** of the upper clamp head **122** at a first twist angle **138** relative to the inner plate **126** of the upper grip **120**.

(30) The lower arm **104** includes a lower grip **140** and a lower clamp head **142**. The lower grip **140**

includes an outer wall **144** and an inner plate **146** (FIG. 6). The lower clamp head **142** includes a bottom **148**, two sides **150**, and a lower clamp plate **152**. The lower clamp plate **152** has a lower hair contact surface **154** that extends between the sides **150** of the lower clamp head **142** at a second twist angle **156** relative to the inner plate **146** of the lower grip **140**.

(31) Moreover, as seen in FIGS. 6-10, the first twist angle **138** and the second twist angle **156** vary along a longitudinal axis **174** of the hair styler **100**. For example, the first twist angle **138** and the second twist angle **156** are different at the longitudinal position shown in FIG. 9 and the longitudinal position shown in FIG. 10. Accordingly, the upper hair contact surface **136** and the lower hair contact surface **154** are twisted about the longitudinal axis **174** and curve in multiple directions.

(32) Referring to FIGS. 6-8, the hair styler **100** also includes teeth **162** extending along edges **160** of the upper hair contact surface **136** and the lower hair contact surface **154**. In the illustrated embodiment, the teeth **162** extend downward from the sides **132** of the upper clamp head **122** and upward from the sides **150** of the lower clamp head **142** and the teeth **162** are configured to engage each other when the hair styler **100** is in the closed position. In addition, the teeth **162** may engage hair positioned between the arms **102**, **104** during the hair styling operation and may comb the hair as the hair styler **100** is moved along the hair. In the illustrated embodiment, the upper clamp head **122** and the lower clamp head **142** each include a single row of teeth **162**. In other embodiments, the upper clamp head **122** and/or the lower clamp head **142** may include two or more rows of teeth **162**.

(33) The hair styler **100** includes an upper end grip **170** attached to the upper clamp head **122** and a lower end grip **172** attached to the lower clamp head **142**. The upper end grip **170** and the lower end grip **172** are on ends of the upper arm **102** and the lower arm **104**, respectively, opposite the upper grip **120** and the lower grip **140**. Accordingly, the user is able to grip the ends of the clamp heads **122**, **142** during a hair styling operation. Gripping the upper and lower end grips **172** allow a user to clamp the styler **100** tighter on the hair and guide the styler along the hair during the hair styling operation.

(34) Referring to FIGS. 9 and 10, the hair styler **100** includes at least one heating assembly **164** enclosed within the upper clamp head **122** and the lower clamp head **142**. The heating assembly **164** is configured to deliver heat to hair throughout the upper and lower hair contact surfaces **136**, **154**. For example, the heating assembly **164** includes heating elements **166** connected to the clamp plates **134**, **152** and to electric control devices for regulating the temperature of the heating assembly. The upper and lower hair contact surfaces **136**, **154** may be heating surfaces of the clamp plates **134**, **152** and be configured to deliver heat to the hair. For example, in the illustrated embodiment, the heating elements **166** extend along the longitudinal axis and are mounted to the clamp plates **134**, **152** at an angle relative to the upper and lower hair contact surfaces **136**, **154**. Accordingly, the heating elements **166** are not twisted like the upper and lower hair contact surfaces **136**, **154** and are able to have a standardized shape. In other embodiments, the hair styler **100** may include any suitable heating assembly.

(35) In some embodiments, the hair styler **100** may include a first insulation layer positioned between the upper hair contact surface **136** and the top **130** and a second insulation layer positioned between the lower hair contact surface **154** and the bottom **148**. The insulation layers may allow the upper and lower clamp heads **122**, **142** to maintain a lower temperature than the hair contact surfaces **136**, **154** when the hair contact surfaces are heated. The insulation layers may include a cold gel, heat insulation foam, and/or any other suitable insulation material.

(36) Referring to FIGS. 11-14, a hair styler of yet another suitable embodiment of the present disclosure is generally indicated by the reference number **200**. The illustrated hair styler **200** comprises a pair of arms (namely, an upper or first arm indicated at **202** and a lower or second arm indicated at **204**), a hinge **206** by which the arms **202**, **204** are connected together, and a power cord **208** (partially illustrated in FIGS. 11 and 14). The hinge **206** is configured to facilitate opening and

closing the arms **202**, **204** relative to one another. The arm **204** is provided with a user interface **210** such as a touch screen including a power button (or “ON/OFF” switch), a temperature increase button (or “UP” switch), a temperature decrease button (or “DOWN” switch), and a graphical user interface (e.g., an LCD display). In addition, the arm **202** may include a lock **212** to lock the styler **200** in the opened or closed position. Suitably, in other embodiments, the hair styler **200** may be configured in any manner that facilitates user operation of the hair styler **200** as set forth herein (e.g., the hair styler **200** may have any suitable number of buttons and displays arranged in any suitable manner).

(37) The upper arm **202** includes an upper grip **220** and an upper clamp head **222**. The upper grip **220** includes an outer wall **224** and an inner wall **226** (FIG. **14**). The upper clamp head **222** includes a top **228**, two sides **230**, and an upper clamp plate **232**. The upper clamp plate **232** has an upper hair contact surface **234** that extends between the sides **230** of the upper clamp head **222** at a first twist angle **236** relative to the inner wall **226** of the upper grip **220**.

(38) The lower arm **204** includes a lower grip **238** and a lower clamp head **240**. The lower grip **238** includes an outer wall **242** and an inner wall **244** (FIG. **13**). The lower clamp head **240** includes a bottom **246**, two sides **248**, and a lower clamp plate **250**. The lower clamp plate **250** has a lower hair contact surface **252** that extends between the sides **248** of the lower clamp head **240** at a second twist angle **254** relative to the inner wall **244** of the lower grip **238**.

(39) Referring to FIGS. **13** and **14**, edges **256** of the upper and lower hair contact surfaces **234**, **252** and the sides **230**, **248** of the upper and lower clamp heads **222**, **240** are free of any teeth or other surface features and are substantially smooth. Moreover, the entire upper and lower hair contact surfaces **234**, **252** between the sides **230**, **248** are substantially smooth and free of surface features. As a result, the configuration of the upper and lower hair contact surfaces **234**, **252** may reduce potential damage to delicate or fine hair during a styling operation.

(40) The hair styler **200** includes an upper end grip **258** attached to the upper clamp head **222** and a lower end grip **260** attached to the lower clamp head **240**. The upper end grip **258** and the lower end grip **260** are located opposite the upper grip **220** and the lower grip **238** on ends of the upper arm **202** and the lower arm **204**. Accordingly, the user is able to grip opposite ends of the upper arm **202** and the lower arm **204** of the hair styler **200** for clamping the upper and lower clamp heads **222**, **240** onto hair and guiding the styler during a hair styling operation.

(41) Embodiments of the hair styler described herein allow a user to curl or straighten hair using a single hair styler. In addition, the twisted configuration of the styler allows the user to comfortably grasp and manipulate the styler during a range of styling operations. Moreover, the twisted hair contact surfaces may be configured to provide a more efficient styling operation than other stylers. For example, the hair contact surfaces may be configured to deliver heat to hair in desired locations as the hair contacts and is styled using the twisted hair contact surfaces.

(42) When introducing elements of the present disclosure or the preferred embodiment (s) thereof, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(43) As various changes could be made in the above constructions without departing from the scope of the disclosure, it is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A hair styler defining a longitudinal axis, the hair styler comprising: an upper arm including an upper grip and an upper clamp head, the upper clamp head including an upper hair contact surface, wherein the upper hair contact surface is twisted relative to the upper grip; a lower arm including a lower grip and a lower clamp head, the lower clamp head including a lower hair contact surface,

wherein the lower hair contact surface is twisted relative to the lower grip; and a hinge attaching the upper and lower arms together such that the upper and lower arms are openable and closeable relative to one another, wherein the hair styler is configured to receive hair between the upper and lower hair contact surfaces when the hair styler is in a closed position, wherein the upper hair contact surface has a first twist angle and the lower hair contact surface has a second twist angle, wherein the first twist angle and the second twist angle each are in a range of 60 degrees to 120 degrees, wherein the first twist angle of the upper hair contact surface and the second twist angle of the lower hair contact surface are symmetrically twisted about the longitudinal axis of the hair styler in the closed position.

2. The hair styler of claim 1, wherein the upper and lower hair contact surfaces fit together when the hair styler is in the closed position.

3. The hair styler of claim 1, wherein the upper clamp head includes a top and an upper clamp plate, the upper clamp plate including the upper hair contact surface and being angled relative to the upper grip.

4. The hair styler of claim 3, wherein the lower clamp head includes a bottom and a lower clamp plate, the lower clamp plate including the lower hair contact surface and being angled relative to the lower grip.

5. The hair styler of claim 4, wherein the top of the upper clamp head and the bottom of the lower clamp head are curved and form a cylinder.

6. The hair styler of claim 5, further comprising a first insulation layer positioned between the upper hair contact surface and the top of the upper clamp head and a second insulation layer positioned between the lower hair contact surface and the bottom of the lower clamp head.

7. The hair styler of claim 1, further comprising a heating system configured to deliver heat to hair positioned between the upper hair contact surface and the lower hair contact surface.

8. The hair styler of claim 1, wherein the first twist angle and the second twist angle vary along the longitudinal axis.

9. The hair styler of claim 1, wherein the upper and lower hair contact surfaces are smooth and free of any teeth.

10. The hair styler of claim 1, further comprising an upper end grip at an end of at least one of the upper arm and the lower arm opposite the upper and lower grips.

11. A hair styler defining a longitudinal axis, the hair styler comprising: a first arm including a first clamp plate, wherein the first clamp plate is twisted; a second arm including a second clamp plate, wherein the second clamp plate is twisted; a hinge attaching the first and second arms together such that the first and second arms are openable and closeable relative to one another, wherein the hair styler is configured to receive hair between the first and second clamp plates when the hair styler is in a closed position; and a heating system configured to deliver heat to the hair positioned between the first and second clamp plates, wherein the first clamp plate has a first twist angle and the second clamp plate has a second twist angle, wherein the first twist angle and the second twist angle each are in a range of 60 degrees to 120 degrees, wherein the first twist angle of the first clamp plate and the second twist angle of the second clamp plate are symmetrically twisted about the longitudinal axis of the hair styler in the closed position.

12. The hair styler of claim 11, wherein the first clamp plate and the second clamp plate fit together when the hair styler is in the closed position.

13. The hair styler of claim 11, wherein the first arm includes an upper clamp head including a top, wherein the second arm includes a lower clamp head including a bottom, and wherein the top of the upper clamp head and the bottom of the lower clamp head are curved and form a cylinder.

14. The hair styler of claim 11, wherein the first arm further includes a first grip attached to the first clamp plate, the first clamp plate being angled relative to the first grip.

15. The hair styler of claim 14, wherein the second arm includes a second grip attached to the second clamp plate, the second clamp plate being angled relative to the second grip.

16. The hair styler of claim 11, wherein the first clamp plate and the second clamp plate are free of any teeth.
17. A hair styler defining a longitudinal axis, the hair styler comprising: a first clamp head including a first hair contact surface, wherein the first hair contact surface is twisted; a second clamp head including a second hair contact surface, wherein the second hair contact surface is twisted; a handle attached to at least one of the first clamp head and the second clamp head; and a hinge attaching the first and second clamp heads together such that the first and second clamp heads are openable and closeable relative to one another, wherein the hair styler is configured to receive hair between the first and second hair contact surfaces when the hair styler is in a closed position, wherein the first hair contact surface has a first twist angle and the second hair contact surface has a second twist angle, wherein the first twist angle and the second twist angle each are in a range of 60 degrees to 120 degrees, wherein the first twist angle of the first hair contact surface and the second twist angle of the second hair contact surface are symmetrically twisted about the longitudinal axis of the hair styler in the closed position.
18. The hair styler of claim 17, wherein the handle includes a first grip attached to the first clamp head and a second grip attached to the second clamp head.
19. The hair styler of claim 18, wherein the first grip and the second grip each include an inner plate, wherein the inner plates are planar, the first and second hair contact surfaces being twisted relative to the inner plates.
20. The hair styler of claim 19, wherein the first twist angle and the second twist angle vary along the longitudinal axis.
-